

Leyan Wu

catherinewu@stu.pku.edu.cn | +86 136-5581-6312 | CatherineWuleyan.github.io

Education

Peking University, B.S. Candidate in Physics & Economics Expected Jun. 2028

- **Selected Coursework:** Econometrics (100/100), Topics in Quantitative Finance(97/100), Linear Algebra (98.5/100), Complex Analysis (98/100), Calculus (97/100), Data Structures and Algorithms (97/100), Computer Vision & Deep Learning, and Modern Physics(100/100).
- **GPA:** 3.92/4.00; **Ranked 1st of 28 in the major; top 1% in the college (all majors)**

Research Experience & Projects

DP Technology Agent Harness Intern Aug. 2026 – Present

- **Paper2IR:** Built a formal reasoning pipeline that converts papers into structured claims and formal relations; processed **103 papers** and constructed cross-paper reasoning graphs.
- **Agent Harness:** Designed a **pluggable execution framework** with reproducible Runs, Artifacts, Checkpoints, and Findings, supporting failure isolation, recovery, branching, and auditability.
- **Autoresearch:** Explored a closed-loop research agent connecting hypothesis generation, experiments, verification, and evidence updates.

Institute of Condensed Matter Physics, PKU Research Intern Aug. 2025 – Aug. 2026

- **2D Material Transport:** Studied Nb₃Cl₈/graphene heterostructures using surface characterization, low-temperature magnetotransport, and DFT to investigate how **layer-dependent surface polarization affects interfacial coupling**.
- **Results:** Combined temperature-dependent Hall measurements with orbital-overlap analysis to explain the polarization-dependent hybridization.

Open-Source & Community Experience

Silicon-Einstein GitHub Organization LKM & Gaia-IR Contributor Aug. 2026 – Present

- Analyzed large-scale Paper2IR failures and identified issues including missing Claim context and conflicts between framework assumptions and real paper reasoning.

DeepModeling Community Agentic Challenge Co-organizer Aug. 2026 – Present

- Contributed to challenge design, review, evaluation, and community support .
- The challenge series received nearly **20,000 submissions**, covering **30+ foundation models** and **60+ agents/harnesses**.

Technical Skills

- **Programming:** Python (NumPy, pandas, SciPy, Matplotlib), Bash, Git
- **Statistics:** Linear regression, maximum likelihood, hypothesis testing, stochastic processes.
- **Scientific Computing:** Signal processing, parameter estimation, curve fitting, and DFT with VASP

Honors and Leadership

- National Scholarship (Highest honor for undergraduates)
- Peking University Yanchuang Group Scholarship (Top 10 in college) & Outstanding Student Award
- Peking University Freshman Scholarship, Second Prize
- Director, Department of Academic and Innovation, Peking University Student Union